

AVIDD ASAP: A71EV2A-c069 and QQ01D2VNS2BA-c003 avitag constructs expression and purification

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Date Started: **2024-Jan-22**

Experiment Started:

Projects: **ASAP**

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Expression

Transformed BL21[DE3]RR with target plasmid and pCDF-BirA (strep/kanR)

Grew 10 mL o/n in superbroth + strep/Kan

Used 10 mL to inoculate 1 L FORM-TB + Kan (3 L total)

Grew 4h 37C 250rpm shaking (OD600 = 1)

Grew 18C 180 rpm shaking 1h

Added 1 mL of 0.5 M D-biotin (stock in DMSO)

Grew 18C 250 rpm shaking 1h

Added 1 mL of 0.5 M D-biotin (stock in DMSO)

Added 1mL of 0.1 M IPTG

Grew 18h 18C 250 rpm shaking

Harvested 4000g 12C 20 minutes

Froze pellet -80C (final wcv = 50 g/L)

Formedium TB custom

12 g/L tryptone

24 g/L yeast extract

3.3 g/L Ammonium Sulphate (NH₄)₂SO₄

6.8g /L Potassium Dihydrogen Phosphate KH₂PO₄

7.1 g/L DiSodium Hydrogen Phosphate Na₂HPO₄ 7.1

0.15 g/L Magnesium Sulphate MgSO₄

0.03 g/L Trace Elements

55.85 g/L in MilliQ water added 20 mL 50 % glycerol

Autoclave

Add 1 mL 10% Antifoam 204 and Antibiotic

Add 20 mL 50 % Glycerol

Add 10 mL 20% Lactose

Add 1 mL 50% Glucose

Purification

Dissolve pellet in 3 mL per g using Lysis buffer (10 mM HEPES pH 7.5, 500 mM NaCl, 5 % glycerol, 0.5 mM TCEP, 1 % TX-100, 30 mM Imidazole, 0.5 mg/mL lysozyme, 0.01 mg/mL benzonase, 0.5 mM D-Biotin)

After dissolving (1h RT stirring)

Freeze -80C o/n 45 mL aliquots

Thaw RT water bath 1h

Centrifuge 4816 g 2h 4C
 Pass over 4 x His Gravitrap columns
 Wash 3 x 20 mL Wash buffer (10 mM HEPES pH 7.5, 500 mM NaCl, 5 % glycerol, 0.5 mM TCEP, 30 mM Imidazole)
 Elute 2.5 mL Elution Buffer (10 mM HEPES pH 7.5, 500 mM NaCl, 5 % glycerol, 0.5 mM TCEP, 500 mM Imidazole)
 Desalt eluate on PD-10 using Base Buffer Mobile phase
 Measure A280 and for every 10 mg of target add 1 mg of TEV protease
 Store o/n in fridge
 Run over HisGravitrap column to remove protease, tag and uncleaved protein
 Concentrate to 10 mg/mL using 10,000 MWCO concentrator
 Do SEC using 5 mL aliquots (approx. 50 mg) on 125 mL Superdex 200 pg (no 75 available) using 10 mM HEPES pH 7.5, 500 mM NaCl, 5 % glycerol, 0.5 mM TCEP as mobile phase
 Pool peak fractions

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A71EV2A-c069 / A71EV2A-k039 / A71EV2A-e050 / A71EV2A-p014

Yield 5.2 mL at 1.8 mg/mL (9.4 mg total)

MSGAIYVGNRYVNRHLATHNDWANLVWEDSSRDLLVSSTTAQGCDTIARCDQCQTGVYYCSSRRKHYPVSFSKPSLIFVEASEYY
 PARYQSHMLAVGHSEPGDCGGILRCQHGVVGIVSTGGNGLVGFADVRDLLWLDEEAMEQH HHHHHHGLNDIFEAQKIEWHE

Number of amino acids: 166

Molecular weight: 18640.77 + biotin (226 Da) = 18866.77

Theoretical pI: 5.84

Extinction coefficients:

Extinction coefficients are in units of $M^{-1} \text{ cm}^{-1}$, at 280 nm measured in water.

Ext. coefficient 34295

Abs 0.1% (=1 g/l) 1.840, assuming all pairs of Cys residues form cystines

Ext. coefficient 33920

Abs 0.1% (=1 g/l) 1.820, assuming all Cys residues are reduced

QQ01D2VNS2BA-c003 / QQ01D2VNS2BA-k006 / QQ01D2VNS2BA-e007 / QQ01D2VNS2BA-p008

Yield = 5 ml at 11.2 mg /mL (55 mg total)

MHHHHHHSSGASWSHPQFEKGGSGGGSGGSAWSHPQFEKSGVDLG TENLYFQSG LNDIFEAQKIEWHEMADLELERAAD
 VKWEDQAEISGSSPILSITISEDGMSIKNEEEEQTLGGGGSGGGGAGVLWDVPSPPPMGKAELEDGAYRIKQKGILGYSQIGAG
 VYKEGTFHTMWHVTRGAVLMHKGKRIEPSWADVKKDLISYGGGWKLEG EWKEGEEVQVLALEPGKNPRAVQTKPGLFKTNAG
 TIGAVSLDFSPGTS GSPIIDKKGKVVG LYGNGVVTRSGAYVSAIAQTEKSIEDNPEIEDDIFRK

SGLNDIFEAQKIEWHEMADLELERAADV KWEDQAEISGSSPILSITISEDGMSIKNEEEEQTLGGGGSGGGGAGVLWDVPSPPP

MGKAELEDGAYRIKQKGILGYSQIGAGVYKEGTFHTMWHVTRGAVLMHKGKRIEPSWADVKKDLISYGGGWKLEGEWKEGEE
VQVLALEPGKNPRAVQTKPGLFKNAGTIGAVSLDFSPGTSGSPIIDKKGKVVGLYGNGVVTRSGAYVSAIAQTEKSIEDNPEIED
DIFRK

Number of amino acids: 258

Molecular weight: 27660.01

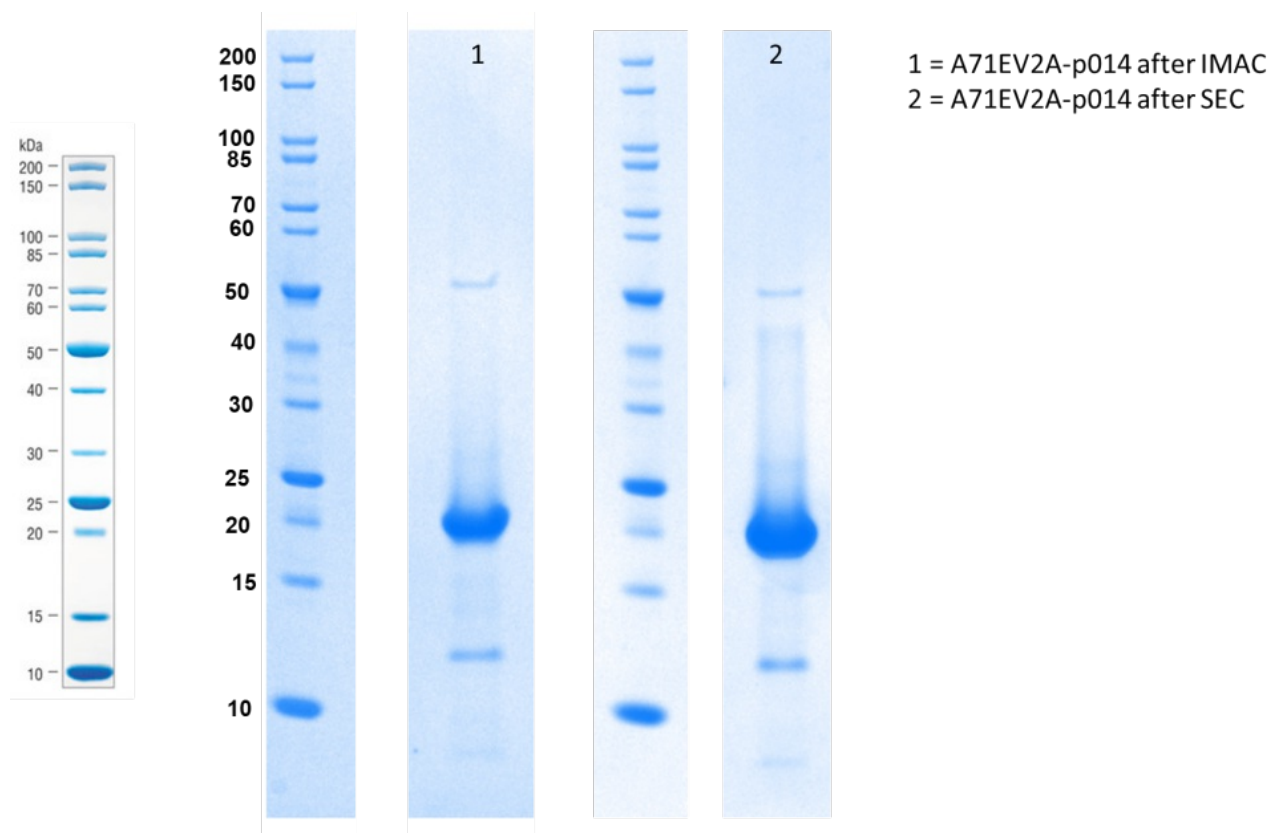
Theoretical pI: 4.92

Extinction coefficients:

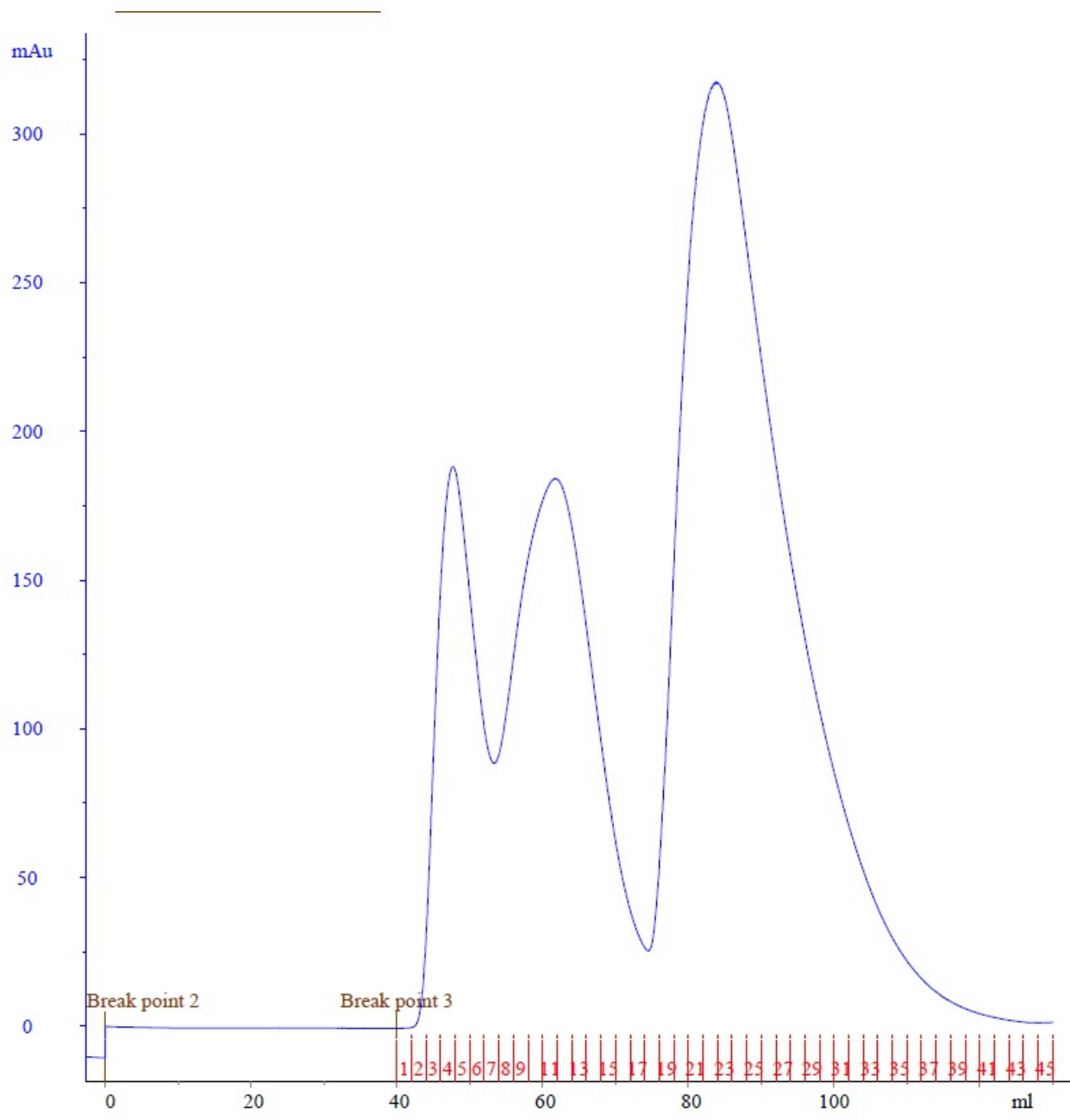
Extinction coefficients are in units of $M^{-1} \text{ cm}^{-1}$, at 280 nm measured in water.

Ext. coefficient 47440
Abs 0.1% (=1 g/l) 1.715

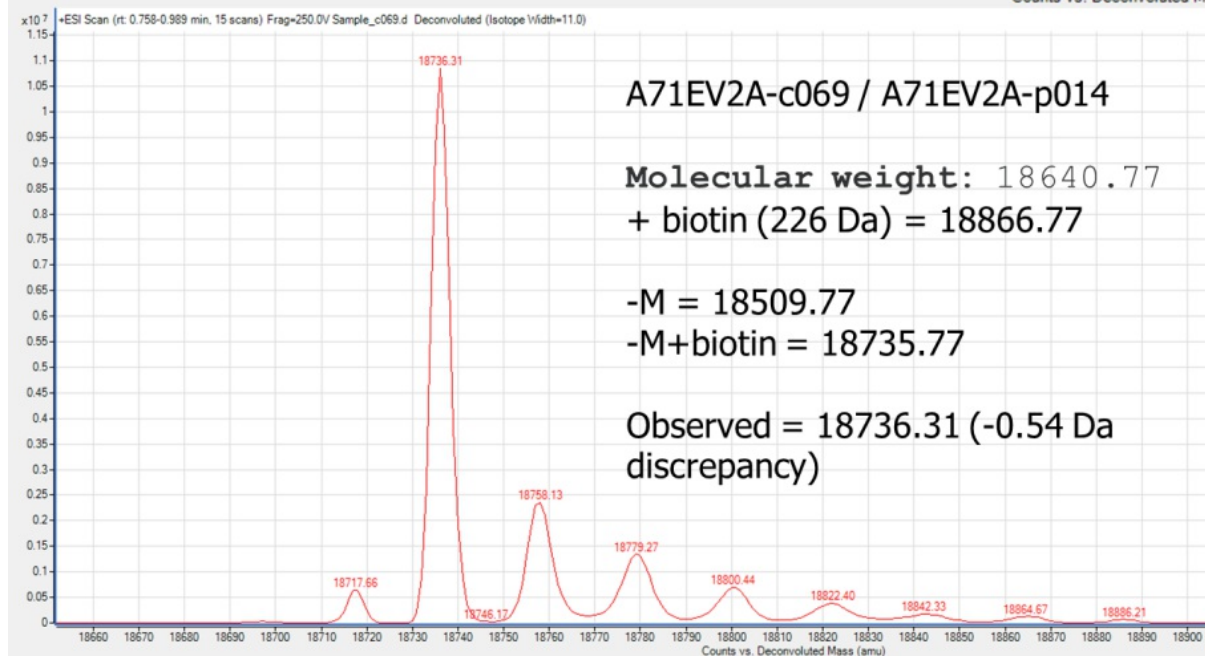
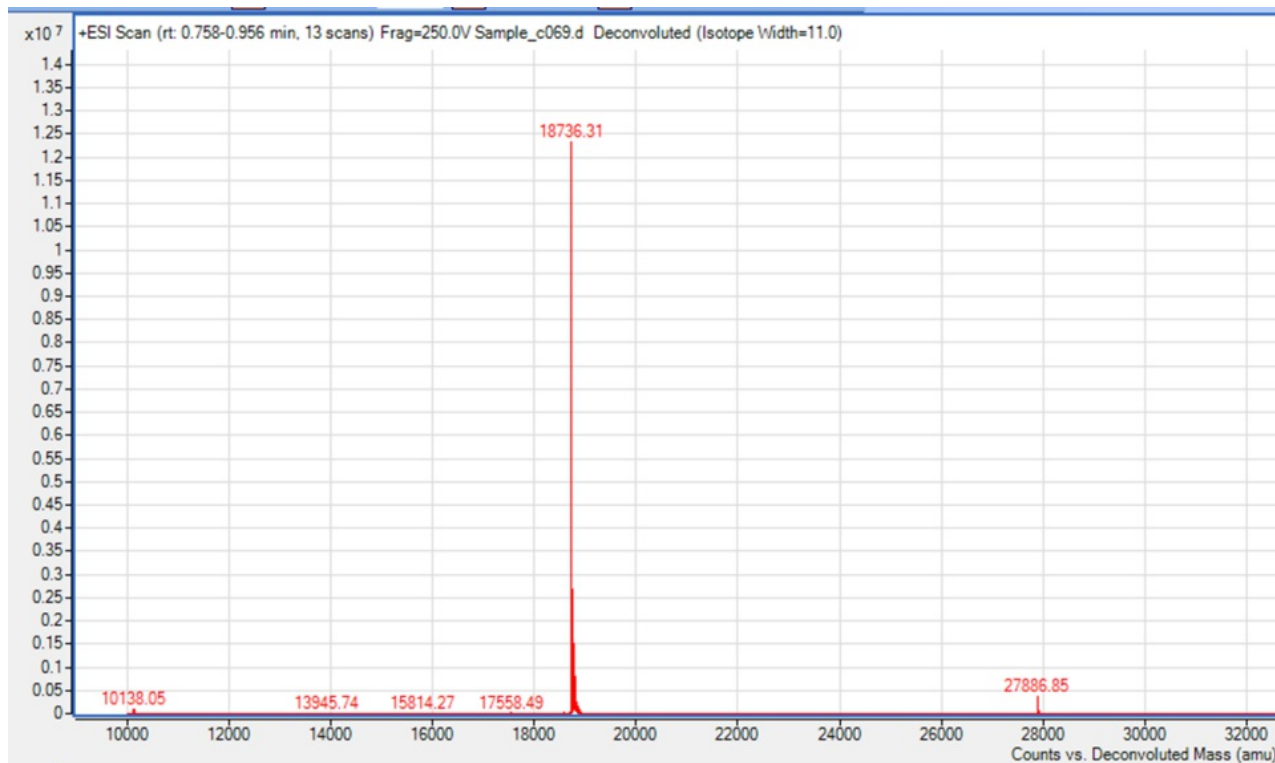
A71EV2A-p014 SDS-PAGE



A71EV2A-c069 SEC profile on 125 mL Superdex 75 pg



A71EV2A-c069 Mass spec

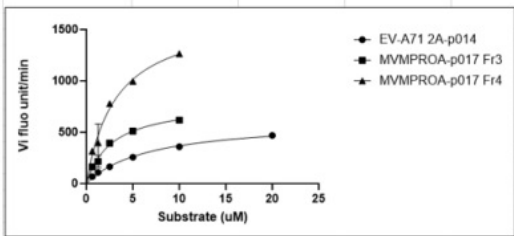


Charline QC of activity

Kinetic: EVA71-2A-p014 MVMPROA-p017 fr3 &4
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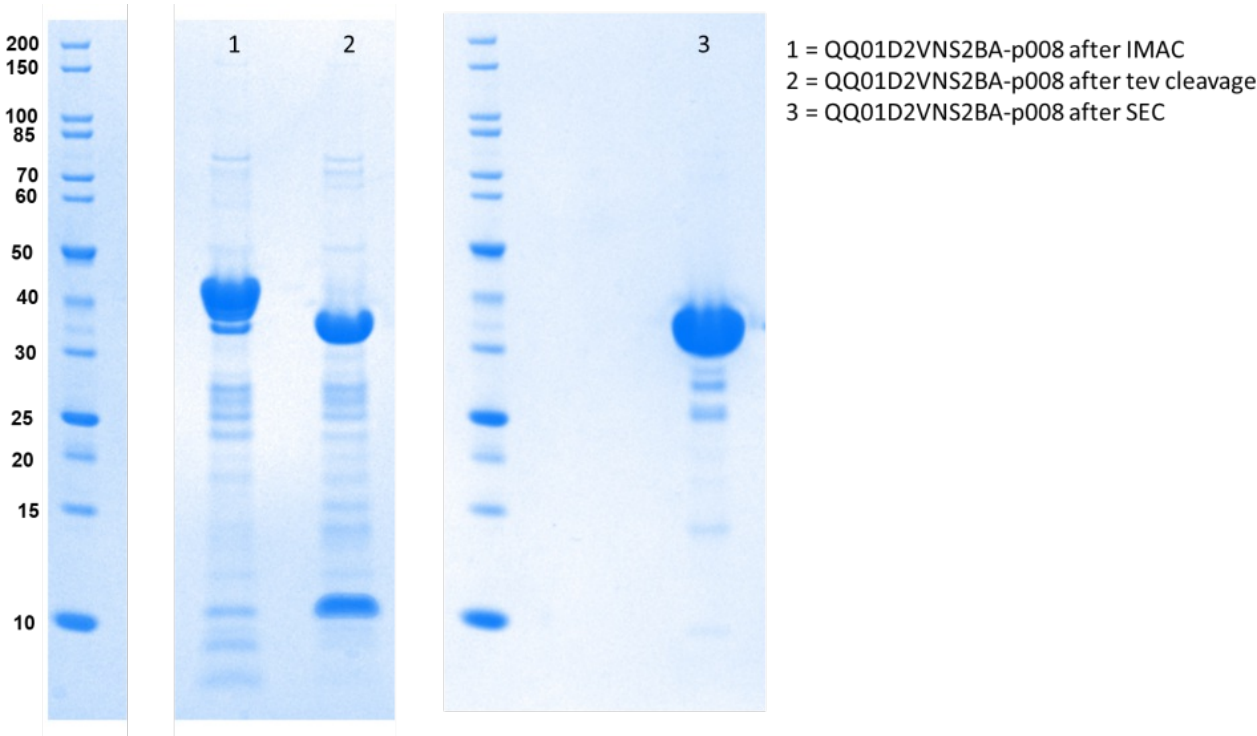
A71EV2A-p007:
Km= 7.99uM
Vmax = 345 fluo unit/min

A71EV2A-p014:
Km=6.732 uM
Vmax = 616.7 fluo unit/min

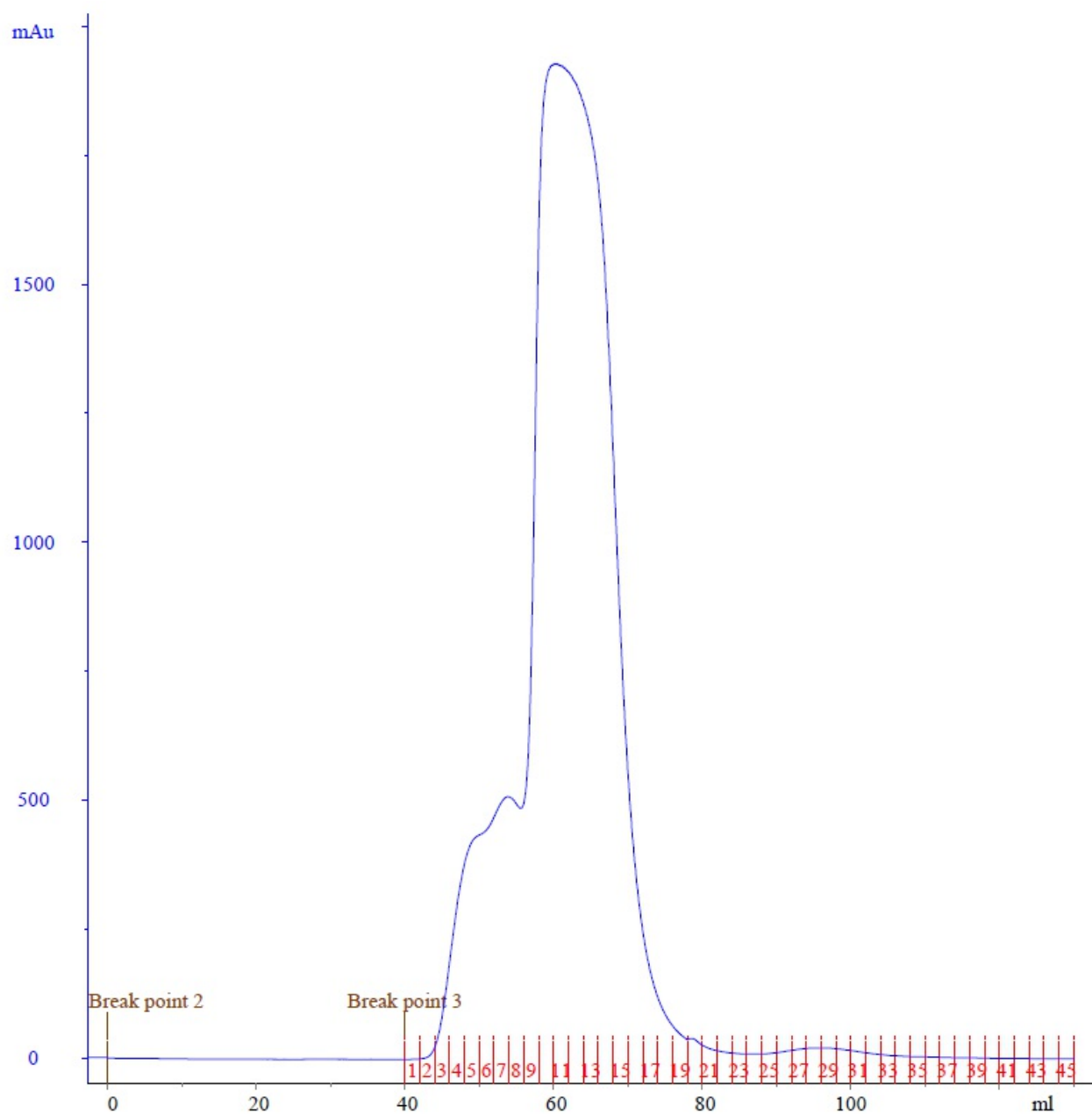


	EV-A71 2A-p014	MVMPROA-p017 Fr3	MVMPROA-p017 Fr4
Michaelis-Menten			
Best-fit values			
Vmax	616.7	794.1	1665 Fluo Unit/min
	0.20574491	0.187119656	0.193059211 uM/min
Km	6.732	2.774	3.203 uM
95% CI (profile likelihood)			
Vmax	555.6 to 691.8	702.8 to 908.5	1401 to 2036
Km	5.216 to 8.768	2.007 to 3.855	2.059 to 5.080
Goodness of Fit			
Degrees of Freedom	10	8	8
R squared	0.9872	0.9756	0.9596
Sum of Squares	3083	7336	52914
Sy x	17.56	30.28	81.33
Constraints			
Km	Km > 0	Km > 0	Km > 0
Number of points			
# of X values	12	12	12
# Y values analyzed	12	10	10
# Y values analyzed	12	12	12

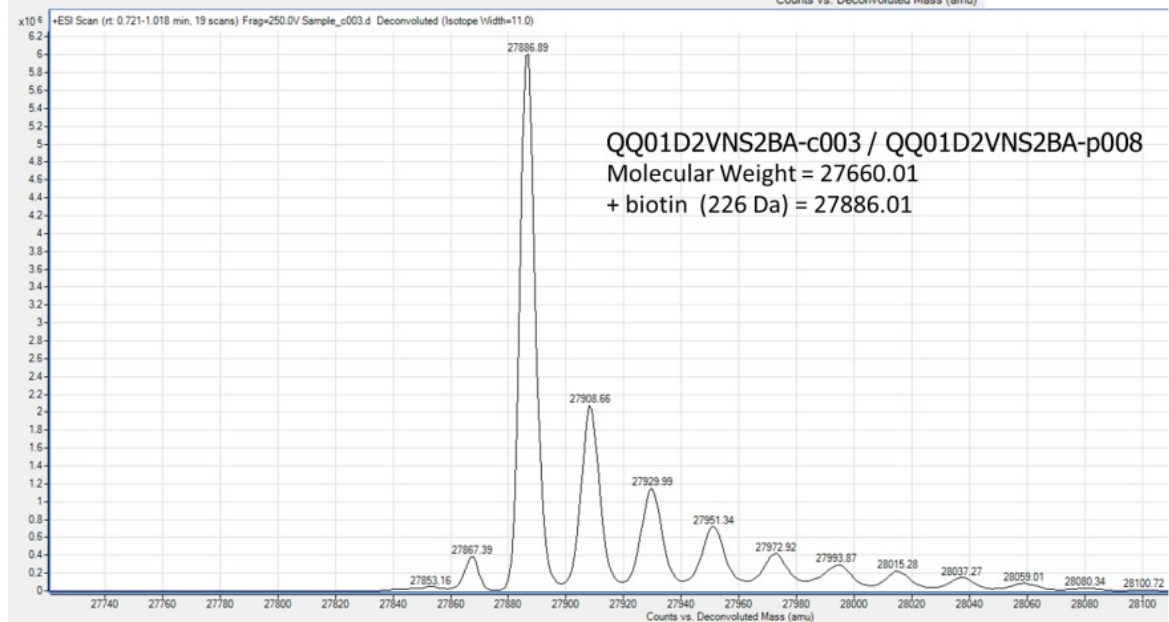
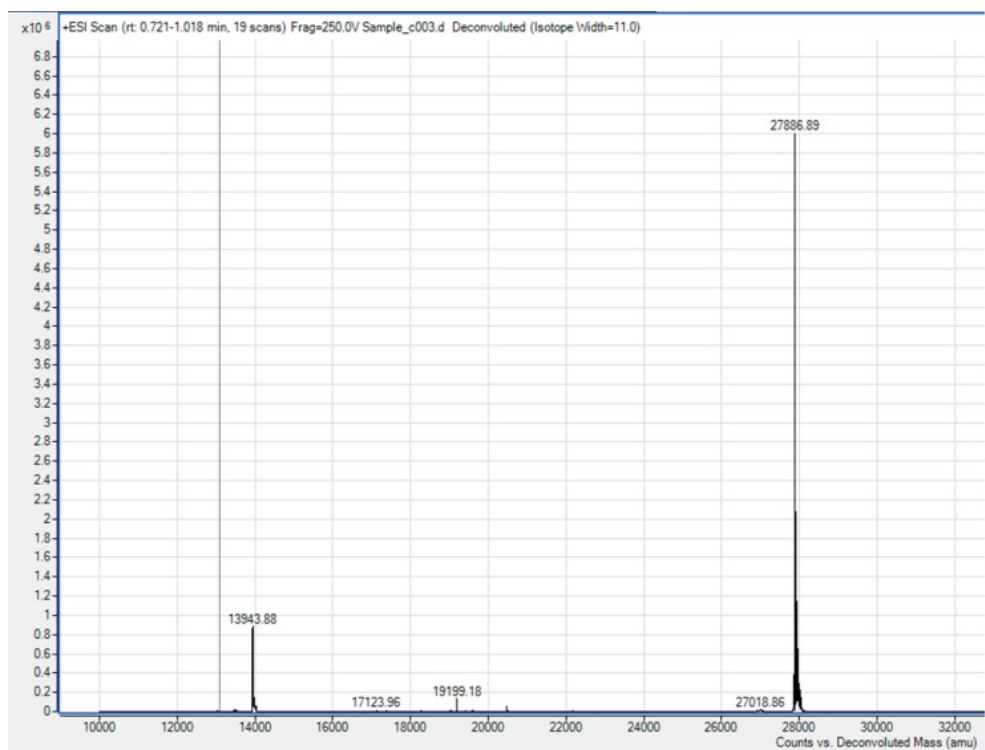
QQ01D2VNS2BA-p008 SDS-PAGE



QQ01D2VNS2BA-c003 SEC profile on 125 mL Superdex 75 pg



QQ01D2VNS2BA-c003 Mass spec



Charline QC of activity

Kinetic: QQ01D2NS2B3-p008 Km Vmax
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- QQ01D2VNS2BA-p008:
 - [E] 0.5uM FAC, [Stock] 335uM
 - [S] 100uM top FAC

QQD2VNS2BB-p001 (no biotin):

Km = 73.42uM

Vmax = 8520 fluo unit/min

QQD2VNS2BB-p008 (+ biotin):

Km = 55.13uM

Vmax = 16938 fluo unit/min

